

THE STARSHIP GALILEO
DEEP SPACE TRAVEL AND COLONIZATION

THE GEMINI PROTOCOLS · TOPICAL BOOK — LIVING VOLUME

THE MEDICAL BOOK



The fleet's doctors and medics. Growing one
section per medical phase.

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THE MEDICAL BOOK — CREW HEALTH & CASUALTY RESPONSE

Gemini Protocols Medical Suite: Phases 163/164 (mass casualty), 99/100 (bone protocol), 88 (atmosphere), 80 (gym), 12 (quarantine), 58/159 (the growing crew), 268 (the gravity prescription — added 12 August 2026), 290 (the G ring — added 18 August 2026)

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What this book is: the fleet's medical doctrine in plain English — the emergency response kit, the daily health systems, and the biological law. It is a living volume: new medical phases get added as they are designed. Version 1, 29 July 2026. Version 2, 12 August 2026 — the Ten-Ring Stair (Phase 268, the ship's gravity) joins as Section 8. Version 3, 18 August 2026 — the G Ring (Phase 290, gravity medicine) joins as Section 10.

SECTION 1 — THE ONE RULE OF SPACE MEDICINE

On Earth, medicine leans on gravity: blood pools, drips fall, weight presses, drains drain. In space none of that is true. Every system in this book replaces a gravity assumption with a physical one:

- **Blood follows Phase 145's water law.** It domes, clings, and floats. Seal it at the skin or breathe it.
- **Pressure must be applied, never assumed.** Snaps, straps, elastic, and ceramic weight — the kit carries all four because gravity carries none.
- **Seconds games are won by removing decisions.** Every package opens one way, every band fits one motion, every protocol is four verbs long.
- **The ship is patient zero.** In a mass-casualty event, hull and life support outrank every casualty — because every casualty's oxygen comes from them.

SECTION 2 — MASS CASUALTY RESPONSE (PHASE 163)

The doctrine: on a ship-wide casualty event, 50% of AI holds the ship (repair, life support, defense); 40% of non-critical service AI runs first response. They are not doctors — they run four verbs: reach, seal, oxygenate, compress.

The pods: six per segregated walkway section, three per side, ceiling-mounted, 8 feet long by about 2.5 feet high. Oxygen and basic supplies racked at one end. Lid seals against smoke.

The board: rigid ambulance-grade slide board, scissor-level at both ends, corrugated underside, rotates up-over-down to the floor keeping the casualty horizontal.

The workflow — treat, seal, leave:

- 1. Stop the bleeding on the spot — field dressing, as in battle.

- 2. Seal the casualty in the pod, air on.
- 3. Relocate out of danger zones if required — the board keeps them horizontal.
- 4. Leave them for the medics and move to the next person. Field dressing is triage, not surgery.

The lone-casualty rule: no corridor design assumes help is coming. If a section is in AI lockdown — or the units are damaged — one injured crew member, alone, one arm broken, must still reach everything. So:

- **Bulkhead wall stations** — both faces of every bulkhead door: a matching pod medical kit, several hand-held oxygen breather bottles (mask on the end), and a CO2 extinguisher beside them. The breather hangs with the extinguisher on purpose: CO2 kills flame by stealing the oxygen, so the tool and its own antidote are one grab apart.
- **Personal pod kits** — every pod carries a smaller medical kit sized to its individual (straps packed to their measurements), plus an extinguisher and a hand-held oxygen breather racked next to the wall repair gun. Fire, breach, breathing, bleeding — one grab point per person.
- **Tether gauntlets (Phase 165)** — wrist-mounted Kevlar line throwers racked on the bridge and in every personal and work pod: for reaching a trapped person after an event, crossing a violently moving deck during a breach, or pulling supplies through an aperture too small for a body. The line goes where the responder cannot.
- **Vitals watches (Phase 166)** — every crew member wears a life-systems watch (heart rate, blood oxygen, rhythm, falls) with two channels: ship telemetry aboard, and a direct link to AI — critical on planetary excursions, where the ship may be over the horizon and the AI unit beside you is the only responder inside the golden hour. When the 50/40 doctrine fires, response order is computed from live vitals, not guessed from silence.

SECTION 3 — THE CASUALTY KIT (PHASE 164)

Where it lives: racked in the pod ends. Everything opens one-handed, gloved, or by AI.

- **Snap bands:** slap-band spring compression, cutlery-drawer racked from thigh to wrist. One motion, even pressure.
- **Squeeze-pop dressings:** burst the air bag, oversized dressing, soft silicone edge seals to bare skin — blood containment is oxygen containment. Body-hair exfoliation regime recommended (head excluded).
- **Apply:** patch on wound, snap band around it. Two motions.
- **Chest, in gravity:** ceramic pie-weight bean bags — the baking beads, pressing where you put them.

- **Chest, in zero-G:** thirty stretch bands per side on the pod's inner walls, crossing the body from any angle, hooked to lower rails.
- **Limb loss, arm/wrist:** neoprene socket over the dressed stump, Velcro strap.
- **Limb loss, leg above knee or extreme:** tourniquet, high and tight, immediately. Lose the leg, keep the life.
- **Injections:** anaphylaxis (epinephrine class); **oxytocin** for postpartum hemorrhage — a nine-amino-acid peptide the ship can ferment from farm feedstocks. Tranexamic acid rides along as the Earth-made deep reserve; its petrochemical synthesis chain can't be rebuilt aboard.
- **Responder protection:** goggles and dust masks in every pod — floating blood is your hazard too.

SECTION 4 — THE BONE PROTOCOL (PHASES 99 & 100)

Two halves of one prescription, designed as complements:

- **All day, static (99):** the PEMS floor pulls each boot down at body weight — the skeleton reads Earth gravity in every step. Watch-item: the floor modulates per stride.
- **Thirty minutes, dynamic (100):** the rotating exercise pool — walk laps through chest-deep water pinned to the wall. Bone grows under dynamic strain, not static weight; the pool supplies the spikes.
- **Upper body:** Phase 80 ceiling-rowing flywheels — the gym that also makes power.
- **Why it works:** marrow sensors read compression and strain, not location. Feet pulled down is Earth, as far as your tibia knows — gravity is gravity, pseudo or otherwise.
- **The chewing doctrine (Phase 169):** teeth, gums, and jawbone obey the same law — occlusal load keeps them, paste loses them. Children growing dentition need chew resistance to build jaw and arch. The galley is part of the bone protocol: texture is medicine.
- **The velodrome dividend (Phase 171):** downforce cycles let children and crew pedal under real load in zero-G — daily leg and hip loading delivered as play. The hardest compliance problem in space medicine, solved by recess.
- **The gravity hoodie (Phase 172):** restraints hold the crew down; nothing but axial load pushes gravity through the spine — and the spine is where microgravity does its worst damage (elongation, disc loss, back pain). The fan hoodie presses 1G through the full vertebral chain, worn as clothing, with a tilt interlock so it only pushes in load posture. Jaw, legs, spine — the bone program is now worn, ridden, and chewed.

SECTION 5 — ATMOSPHERE AS MEDICINE (PHASE 88)

- The breathing reserve is solid Nitropril prills — nitrogen stored as rock, unable to leak, flashed to breathing gas on demand in measured batches. The audit's correction stands: flash above ~300C for nitrogen, with the scrub guard against the low-temperature (laughing gas) pathway.
- Kidney stones and bone loss share a cause — calcium leaving bone. Stop the resorption (Section 4) and the bloodstream stays clean.

SECTION 6 — QUARANTINE LAW (PHASE 12)

- Anything biological from off-world passes agar-plated barriers and Level-5 incineration locks. No exceptions, no heroic samples, no shortcuts — the crew's immune systems are the frontier's softest hardware.

SECTION 7 — THE GROWING CREW (PHASES 58 & 159)

- **The demographic law (58):** births capped at the 20-birth ceiling, pediatric pods standardized, no corporate custody of children — a century-ship's population is engineered like its water.
- **The nursery (159):** vibrational beds for germination — plant and crew agriculture alike start life shaken, not still.

SECTION 8 — THE GRAVITY PRESCRIPTION: THE TEN-RING STAIR (PHASE 268)

The wound, named plainly: SANS, bone loss, muscle loss and fluid shift are not four problems — they are one. The body parked in freefall lets every column in it drift upward: the blood rises, the eyeballs flatten, the bones unload, and — the Author's extension, seated 12 August 2026 — if the blood rises, the intestinal contents and the large bowel's fluid swap suffer the same issues. The gut rides with the blood.

The machine: ten rings side by side, each 6 m high and a metre wide, spinning at graduated speeds. You step on in zero-G, wait three seconds, walk to the next ring, wait three seconds, repeat — the variance is so small you barely need to step forward — until the last ring, where the rim turns at 1G and you can walk, stand, take a chair and read a book, or hold the long meeting on the wide wheel. The prescription is TIME, not exercise.

THE PRESCRIPTION (as amended 12 August 2026): 20 minutes twice daily, the two sessions at least 6 hours apart, preferably 8 to 12 — two spaced pulls of the fluid column, so the gut column re-settles twice a day. This supersedes the original 30 minutes once daily (seated 9 August 2026, preserved in the master). Fleet fit: 10–20 units per ship — note the per-person demand rose from 30 to 40 minutes daily, so the count runs a third tighter.

The honest caveat, carried with the medicine: 17.3 RPM on the last ring is above the unadapted-comfort line — a 1 m/s head turn there pulls ~0.37 g of Coriolis sideways. First-fortnight users will feel head

turns. The design's own countermeasure: the graduated 3-second dwells ARE the adaptation ramp — the stair teaches the vestibular system on the way up.

SECTION 9 — OPEN ITEMS & WHAT'S COMING

- **CLOSED — childbirth injection: oxytocin.** Archer's selection rule: whichever is easier to synthesize in space. Oxytocin — nine amino acids, fermentable in a bioreactor — beats tranexamic acid — petrochemical precursor chain, high-pressure hydrogenation — on a ship with farms and no refinery. TXA stocked from Earth as reserve.
- More phases are expected in this book — Archer has flagged further medical designs. Each arrives as a new section with its own physics audit and Origin of Thought.

SECTION 10 — THE G RING: GRAVITY DRIVE QUARTERS & THE GRAVITATIONAL HEALTH MANDATE (PHASE 290)

The mandate, stated plainly: the ship is zero-G by baseline, and the body was not built for a lifetime of it. The ten-ring stair (Section 8) is the prescription; the G ring is the ward — a full-gravity deck where the fleet's medicine, pregnancies, recoveries and childhoods happen under a real sky-load. The author's dictation is seated verbatim in the master's Phase 290; this section is the working doctrine.

- **The machine:** a 50m-diameter × 20m-wide ring spun for 1G at the deck — about 6 RPM, 15.7 m/s rim speed — fed by the Phase 268 g-stairs: you step up the stair and walk onto the deck. Deck area $\pi \times 50\text{m} \times 20\text{m} = 3,141.6 \text{ m}^2$ — the author's stated '3/4 of an acre', verified within four percent.
- **The 8-hour law:** 8 hours per day on deck for everyone inside, minimum and maximum approximations his — the daily dose that lets newborns and children transition to ship life instead of merely surviving it.
- **The childhood law:** children live in the ring until age 9, leaving only for school, events and daily routines inside the 8-hour window. Babies under 2: a maximum of 4 hours out for events, 2 hours otherwise — until the skull closes (the fontanelle line; his under-2 bound aligns with the 9-18 month closure).
- **The medical deck in gravity:** pregnancy quarters, surgery units, recovery and ICU inside the ring — safety for the mother, no floating blood during surgeries, a stable environment for recoveries. Every zero-G workaround in Sections 1-3 still stands for the rest of the ship; inside the ring, Earth rules apply.
- **The failsafe:** if the ring fails, nothing happens — the ship is zero-G by baseline; ring-down is ship-normal. Carried honestly with it: a spinning 50m ring stores rim energy, so containment, overspeed and bearing specifications are owed engineering — a work order, not a veto.

- **The wall law:** the ring's side is a single fixed colour with a step entrance — no view out to the rotation or the shell wall. Looking up at the curving deck is strange enough; it mirrors ship life anyway.
- **The livestock line:** mini G wheels for incubation — eggs and amniotic fluids obey the same gravity law as babies (flown egg experiments — quail on Mir, chicken embryos on Shuttle — showed gravity-sensitive development; the chicken-egg musing has precedent). The chickens-and-livestock breeding assessment is the author's stated pending work. Weight/restraint vests serve grooming and vet checks; the veterinarian seat rides the crew roster with a husbandry qualification requirement.

The honest caveat, carried with the medicine as always: 6 RPM sits above the unadapted-comfort line — adaptation is demonstrated and children adapt fastest, but the spin rate is the author's to tune, and the ring's placement bay is his to rule at layout day. Hull-length increase is pre-authorized only if layout requires it — no length change has been ordered.